

Erdős Problem #538

EP-538

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $r \geq 2$ and suppose that $A \subseteq \{1, \dots, N\}$ is such that, for any m , there are at most r solutions to $m = pa$ where p is prime and $a \in A$. Give the best possible upper bound for $\sum_{n \in A} \frac{1}{n}$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>